Ch 2.2: Assessing Model Accuracy

Lecture 3 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Fri, Jan 16, 2026

Announcements

Last time:

- Ch 2.1, Vocab day!

Announcements:

- First homework due Sunday, 1/25. Covers:
  - Mon 1/12 lecture
  - Weds 1/14 Lecture
  - Today 1/16 Lecture

Covered in this lecture

- Mean Squared Error (regression)
- Train vs Test
- Bias Variance Trade off

Quick review of notation

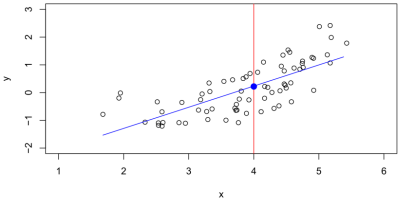
- $X = (X_1, \cdots, X_p)$ number of variables
- Ground truth $Y = f(X) + \varepsilon$
- Approximation $\hat{Y} = \hat{f}(X)$
- Number of data points n
- X_{ij} is jth predictor for observation i

Section 1

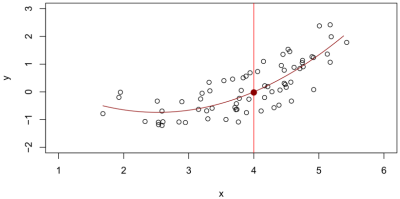
Mean Squared Error

Which is better?

A linear model $\hat{f}_L(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ gives a reasonable fit here



A quadratic model $\hat{f}_Q(X) = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$ fits slightly better.



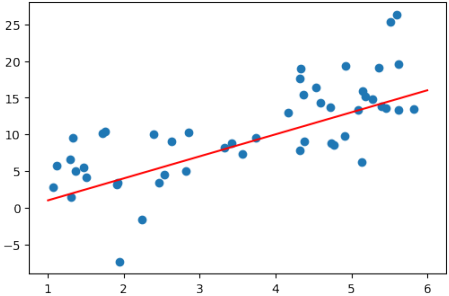
No free lunch

- No best model for all data sets
- So, need to measure quality of a model on a given data set

Mean Squared Error

Error in the regression setting

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$$



Group Work

Given the following data, you decide to use the model

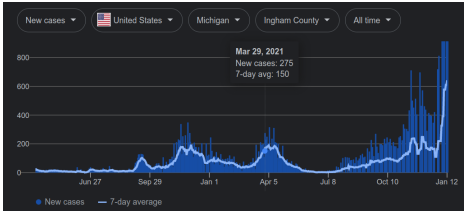
$$\hat{f}(X_1, X_2) = 1 - 3X_1 + 2X_2.$$

What is the MSE?

X_1	X_2	Y
0	7	14
1	-3	-6
5	2	-10
-1	1	7

hat f = 1-3*X_1+2*X_2	Abs val of dif	squared dif
15	1	1
-8	2	4
-10	0	0
6	1	1
		6

Training MSE



We don't care about how well the model does on the data we have..... we want the quality of model on data not seen when training the model

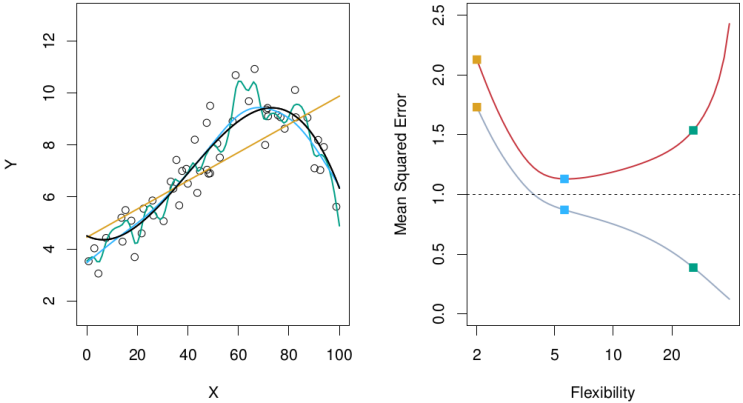
Train vs test

Training set:
The observations
$\{(x_1, y_1), \cdots, (x_n, y_n)\}$ used to get
the estimate $\hat{f}$

Test set:
The observations
$\{(x'_1, y'_1), \cdots, (x'_{n'}, y'_{n'})\}$ used to
compute the average squared
prediction error

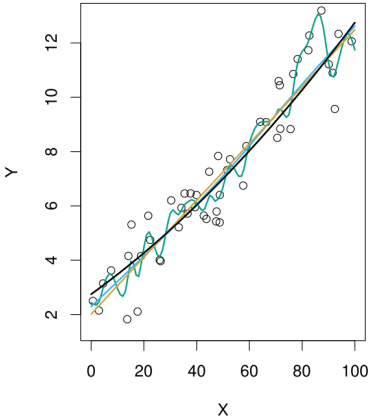
$$\frac{1}{n'} \sum_i (y'_i - \hat{f}(x'_i))^2$$

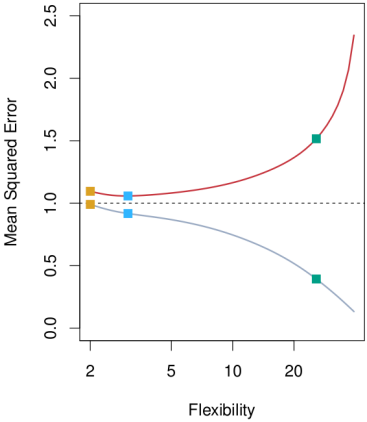
Why not just get the best model for the training data?



- Left: Black line is model for simulation of data, three estimates of f are shown in orange, blue and green curves
- Right side: training MSE in blue/grey, testing in red
- Squares represent the training and test MSEs for the three fits shown in the left-hand panel
- Variance of ε is dashed line
- Point out that training error goes down but test goes up (overfitting)
- Flexibility = degrees of freedom

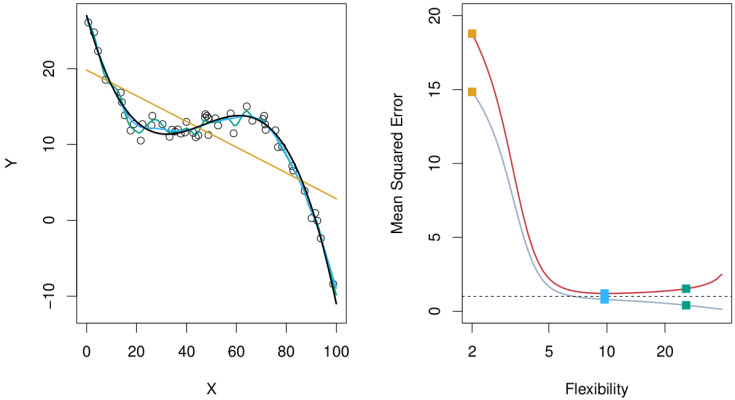
A more linear example





- Truth is linear, so test MSE down only a bit before going up

A more non-linear example



- Similar structure to previous but needs more degrees of freedom before getting a good test

A simple solution: Train/test split

More on this in Ch 5

Section 2

Bias-Variance Trade-Off

Bias-variance

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

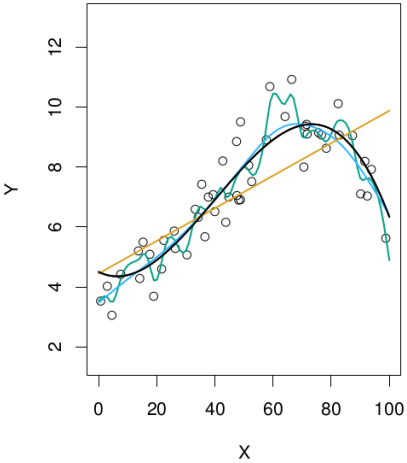
- Proof outside of the scope of class (See Bias-variance tradeoff on Wikipedia)
- $E(y_0 - \hat{f}(x_0))^2$ is expected test MSE at x_0; avg test MSE if repeatedly estimated f with lots of training sets and tested each at x_0
- Computed by averaging $E(y_0 - \hat{f}(x_0))^2$ over all values in the test set

- Eqn says we need an $\hat{f}$ with both low variance and low bias
- Also says error is bounded below by irreduceable error $\mathrm{Var}(\varepsilon)$

Variance

Variance: the amount by which $\hat{f}$ would change if we estimated it using a different training data set.

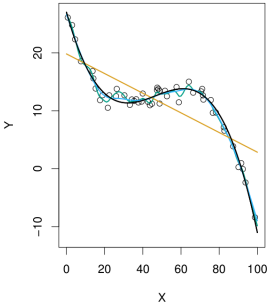
- High variance: small changes in training data result in large changes in $\hat{f}$.
- Example right: green curve more flexible, but also small changes in data set could cause large changes in the computed model.

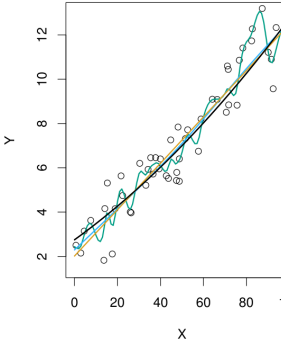


Bias

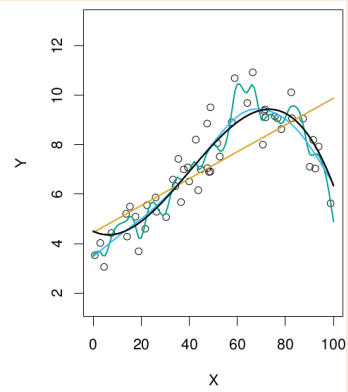
Bias: the error that is introduced by approximating a (complicated) real-life problem by a much simpler model.

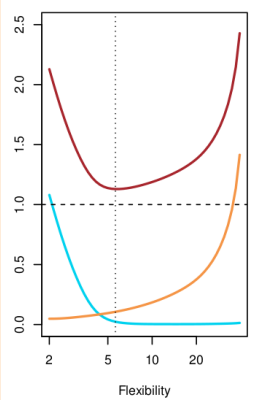
- Example: Linear regression, likely too simple for any real-life problem.
- Figure at left: True f is non-linear, so no good estimate possible with linear regression. (linear regression = high bias)
- Figure at right: True f is linear, so linear regression should do a good job (linear regression = low bias)





Group work



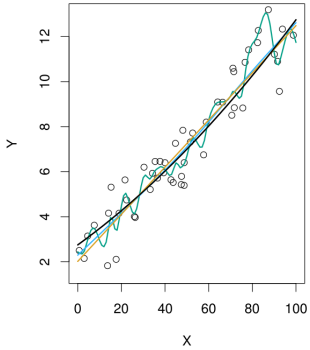


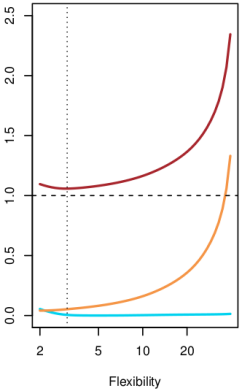
Label the line corresponding to each of the following:

- MSE
- Bias
- Variance of $\hat{f}(x_0)$
- Variance of ε

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Another example





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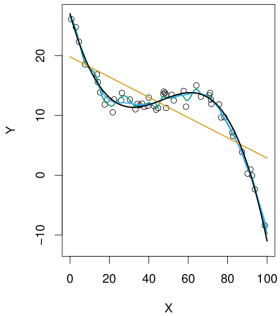
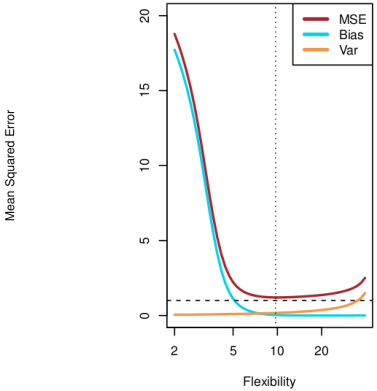
Yet another example





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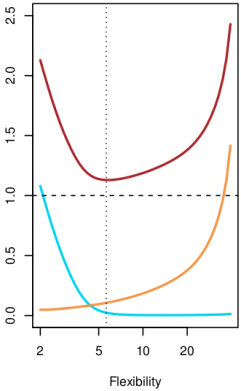
Bias-variance trade off



- More flexible: variance up, bias down.
- Relative rate of change between bias and variance determines whether MSE goes up or down
- Initially, bias goes down faster, so MSE declines
- Eventually little effect on bias but lots of increased variance, so MSE increases

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Group work: coding

See jupyter notebook

Next time

- Wednesday:
  - 3.1 Linear Regression
- Sunday
  - Homework due midnight on www.msu-stt-learning.com

Lec #	Date		Topic	Reading	HW	Pop Quizzes	Notes
1	M	1/12	Intro / Python Review	1			
2	W	1/14	What is statistical learning	2.1		Q1	
3	F	1/16	Assessing Model Accuracy	2.2.1, 2.2.2			
	M	1/19	MLK - No Class				
4	W	1/21	Linear Regression	3.1		Q2	
5	F	1/23	More Linear Regression	3.1	HW #1 Due Sun 1/25		
6	M	1/26	Multi-linear Regression	3.2			
7	W	1/28	Probably More Linear Regression	3.3		Q3	
8	F	1/30	Last of the Linear Regression		HW #2 Due Sun 2/1		
9	M	2/2	Intro to classification, Bayes classifier, KNN classifier	2.2.3			